

Interfacing Dot Matrix LED with Raspberry Pi

SOMNATH BERA

You will find in this article projects for interfacing an 8×8 and a 5×7 dot-matrix display with Raspberry Pi.

Since Raspberry Pi does not have enough GPIOs, we have used its I²C bus to connect to MCP23017 IO expander to have 16 additional GPIOs for connecting to LED dot-matrix display such as 8×8 or 5×7. Pin 3 (SDA) and pin 5 (SCL) are I²C pins on Raspberry Pi board. Some libraries in the form of a GPL Python script, taken from Adfruit.com, are used in this project to manage the extended GPIOs.

First, you will see how a single 8×8 LED matrix is interfaced with GPIOs through MCP23017. Then you can expand the project to interface more dot-matrix displays.

System set up

To use I²C on Raspberry Pi, you need to enable a few things in Raspbian Wheezy operating system as these are not enabled by default. This is a fairly easy process. First you edit the modules file using the command:

```
$ sudo nano /etc/modules
```

and add the following two lines (also refer Fig. 1):

```
i2c-bcm2708
i2c-dev
```

Save the file and exit nano. Next, open the rpi-blacklist.conf file by writing the command (also refer Fig. 2):

```
$ sudo nano /etc/modprobe.d/
rpi-blacklist.conf
```

Comment out the code by putting # symbol at the beginning of the two lines as given below (also refer Fig. 3) and then save and exit.

```
#blacklist spi-bcm2708
#blacklist i2c-bcm2708
```

Now install the necessary software

to manipulate the I²C bus using following commands (also refer Fig. 4):

```
$ sudo apt-get update
$ sudo apt-get install python-usb
i2c-tools
```

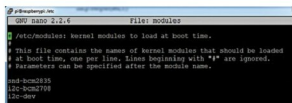


Fig. 1: Adding the lines in /etc/modules



Fig. 2: Editing the rpi-blacklist.conf file

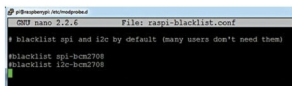


Fig. 3: Commenting the code

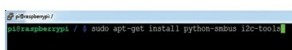


Fig. 4: Install the Python-usb and I²C tools

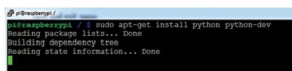


Fig. 5: Installing Python packages



Fig. 6: Extracting the tar file

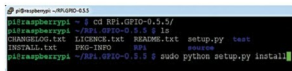


Fig. 7: Installing the setup.py

Once the software is installed, reboot your Raspberry Pi so that these effects take place permanently. Also install Python and python-dev on Raspberry Pi using following commands (also refer Fig. 5):

```
$ sudo apt-get install
python python-dev
```

For interacting with the GPIO through Python, you have to install the GPIO Python libraries. The latest library available is RPi.GPIO-0.5.5.tar.gz. Download the same from the link below:

<http://raspberrypi-python.googlecode.com/files/>

Unzip or extract the tar file by using command (also refer Fig. 6):

```
tar xzf RPi.GPIO-
0.5.5.tar.gz
```

Now change directory to the RPi.GPIO-0.5.5 and write the following command (also refer Fig. 7):

```
$ sudo python setup.py
install
```

The command will install Raspberry Pi GPIO library into Python. Now reboot the Raspberry Pi using command:

```
$ sudo reboot
```

Project 1: Multiplexing of 8×8 dot-matrix display

In this project, you will use two separate files — one to create a template

